

HongKun Tian

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SUMMARY

Backend and distributed-systems engineer with 4+ years on AWS Step Functions, owning production reliability, workflow-platform capabilities, global region launches, deployment automation, and operational tooling for a large-scale serverless service.

EXPERIENCE

Amazon Web Services (AWS)

Software Development Engineer II, AWS Step Functions

Jul 2022 – Present

- Owned end-to-end modernization of Step Functions alarm infrastructure, automating deployments, consolidating 1,016 managed resources to 75, cutting synthesis time from 93 to 26 minutes, and eliminating redundant per-service paging in migrated regions.
- Designed and implemented core backend functionality for state machine versioning and aliases, enabling customers to publish immutable workflow versions and safely shift production traffic.
- Built an SOP-driven LLM agent with reusable tools to apply a complex quota-safety decision tree, reducing manual triage by approximately 80% across 10-20 weekly requests while retaining human review.
- Drove Step Functions feature launches and parity across commercial, China, GovCloud, and isolated AWS Regions, resolving cross-service deployment failures in restricted environments.
- Built throttling, monitoring canaries, and global rollout infrastructure for the ValidateStateMachineDefinition API, enabling customers to catch workflow-definition errors before deployment.
- Delivered JSONata expression validation that prevented invalid runtime references and improved developer diagnostics for a major workflow-language launch.
- Built operational tooling to safely roll out customer quota changes across 168 accounts and 24 AWS Regions and monitor production workflow regressions.
- Delivered CloudFormation resource-provider capabilities for Step Functions, including lifecycle handlers and automated integration tests for infrastructure-as-code workflows.

Software Development Engineer Intern, AWS Step Functions

May 2021 – Aug 2021

- Built reusable React components, workflow visualizations, and cross-service integration tests for AWS Step Functions console integrations.

Ericsson

Software Developer Intern

Jan 2021 – Apr 2021

- Automated Jenkins deployment workflows across VMware and OpenStack environments using Bash and Groovy.

UMAknow

Software Developer Intern

Jun 2019 – Aug 2020

- Built cloud-infrastructure diagram generation, a server-side REST API, responsive UI, and automated acceptance tests for Clouddockit.

EDUCATION

McGill University

Montreal, Canada

Bachelor of Science, Honours Computer Science

Sep 2018 – May 2022

GPA: 4.00 / 4.00; Alan Shaver Scholarship for academic excellence.

SKILLS

- Languages:** Java, Python, TypeScript, JavaScript, C/C++, Bash, SQL
- Infrastructure:** AWS, Step Functions, CloudFormation, CDK, DynamoDB, CloudWatch, EventBridge
- Systems:** Distributed systems, workflow orchestration, backend services, CI/CD, infrastructure automation, service reliability
- Developer Tooling:** API design, integration testing, observability, production debugging
- AI / Tooling:** LLM agents, tool calling, SOP-driven automation